



Name: Cohort/Batch: Date: Score:

WEBASSEMBLY PRACTICE SHEET - 10 CONCEPTUAL Q&A

10 conceptual Q&A Emerging Technologies

TOTAL: 10 QUESTIONS

Q1 What is WebAssembly and what problem in Emerging Technologies does it solve? **Threat Model**

Q6 How does WebAssembly handle reliability and high availability? **Observability**

Q2 What are the core components or concepts in WebAssembly? **Assembly2e**

Q7 What observability does WebAssembly provide out of the box? **the box2**

Q3 How does WebAssembly compare to its main alternatives? **alternatives2e**

Q8 What are common performance bottlenecks in WebAssembly? **Assembly2e**

Q4 How do you get started with WebAssembly for a new project? **project2e**

Q9 What security best practices apply when running WebAssembly? **Testing**

Q5 What configuration options most affect WebAssembly's behavior in production? **SSO / IdP**

Q10 What mistakes do teams commonly make when adopting WebAssembly? **Tuning**



ANSWER KEY & EXPLANATORY SOLUTIONS

Comprehensive verified answers, best-practice configs, security controls & reference

VERIFIED SOLUTIONS

A1 WebAssembly is a tool or platform that

Mitigation

WebAssembly is a tool or platform that automates or simplifies a specific concern in the Emerging Technologies domain, reducing manual effort and operational complexity for engineering teams.

A6 WebAssembly provides HA through leader election, Observability

WebAssembly provides HA through leader election, replication, or stateless horizontal scaling. Deploy at least two instances across availability zones and configure health checks for automatic failover.

A2 WebAssembly has key building blocks that work

Primitives

WebAssembly has key building blocks that work together: a configuration layer defines intent, a runtime layer enforces it, and an API or CLI layer allows operators to manage and observe the system.

A7 WebAssembly exposes metrics (Prometheus endpoint), Information

WebAssembly exposes metrics (Prometheus endpoint or built-in dashboard), structured logs, and optionally distributed traces. Define SLOs on the key metrics and alert before users are affected.

A3 WebAssembly trades off simplicity against feature richness, Missing

WebAssembly trades off simplicity against feature richness differently than its alternatives. Choose WebAssembly when its specific strengths performance, ecosystem, or ease of use align with your team's primary constraints.

A8 Performance bottlenecks typically stem from misconfigured, Performance

Performance bottlenecks typically stem from misconfigured connection pools, large payload sizes, insufficient worker threads, or missing caches. Profile with built-in diagnostics before optimizing.

A4 Install WebAssembly via its package manager or

Gotchas

Install WebAssembly via its package manager or binary release. Follow the quickstart to initialize a project, configure the minimal required settings, and run the default command to verify the setup works end-to-end.

A9 Run WebAssembly with the principle of least

Assessment

Run WebAssembly with the principle of least privilege, enable TLS for all communication, use a secrets manager for credentials, and keep the software patched to the latest stable release.

A5 Production-critical settings typically include concurrency, Configuration

Production-critical settings typically include concurrency limits, timeout values, retry policies, resource limits, and logging levels. Review the official production hardening guide before deploying.

A10 Common pitfalls include skipping the documentation and, Configuration

Common pitfalls include skipping the documentation and learning the API by trial and error, using default configurations in production, lacking a runbook for operational issues, and not testing failover scenarios.